Toxic Comment Classification Project.

Course Report: Deep Learning and Big Data

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Abstract

This project studied the Jigsaw Toxic Comment Classification dataset, a collection of online comments with labels for six forms of toxicity. The analysis showed that the dataset is strongly imbalanced. In the training file, 143,346 of 159,571 comments were clean, while 16,225 comments had at least one toxicity label. That means only 10.17% of training comments were unclean. The project follows a requirement to make two models from scratch and then, with an alternative approach using a pretrained model. The first task was multi-label classification, where one comment can receive more than one toxicity label. The second task was binary classification, where the six original labels were combined into one clean versus unclean decision.

The from-scratch models used a simple but complete PyTorch text classification design. Text was lowercased, simplified to letters, numbers, and spaces, tokenized by words, and converted into integer IDs. The multi-label from-scratch model used a 50,000-word vocabulary, which covered 98.38% of token occurrences in the training data. The transfer-learning models used distilbert-base-uncased through Hugging Face Transformers. DistilBERT is a pretrained language model, so it begins training with general language patterns already learned.

The results support a clear conclusion that DistilBERT performed better overall than the from-scratch neural networks.

1.0 Introduction

Online discussions heavily depend on comments written by users on social media platforms. These comments can be useful, playful, argumentative, or informative, but they can also become hostile or harmful. A large platform that gets millions of written comments a day can not manually filter them for what is considered toxic or not. The Motivation for choosing this data set is to explore how machine learning methods can help identify toxic and harmful comments in online conversations.

The focus of this project is the jigsaw Toxic comment classification data set. There are six possible labels in the data set. In the data set, a comment can have multiple labels at the same time. There is also a binary task to combine all labels into one class named unclean comments, so that comments can be classified as simply clean or unclean. A comment is unclean if at least one of its labels is positive. This second task of binary classification is useful because in the real-world workflow, there is first a question of whether a comment should be reviewed or not.

2.0 Dataset description

This project focuses on toxic comment classification with the use of the Jigsaw Comment Classification dataset. The dataset comes from Kaggle: https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge. The project contains three data files as described below in the table

File	**Rows**	Columns	**Purpose**
train.csv	159,571	8	Training comments with text and six toxicity labels
test.csv	153,164 **(63,978 usable)**	2	Test comments with IDs and comment text without labels
test_labels.csv	153,164 **(63,978 usable)**	7	Labels for the test comments

TABLE 1. LOCAL DATASET FILES USED BY THE PROJECT.

Each target column is binary, with 1 meaning the label is present and 0 meaning it is absent. The labels are not mutually exclusive. A row can have no active labels, one active label, or several active labels. The binary clean/unclean target was created from these six labels by counting how many labels were active for each row. test.csv has **153,164 comments**, but test_labels.csv contains **89,186 rows where the labels are -1**. In the Kaggle dataset, -1 means those rows were **not scored / not usable for evaluation**. Since a confusion matrix needs the true label to know whether each prediction is right or wrong, those rows are removed.

2.1 Dataset quality check

There are no missing values in the loaded files. There are also no duplicate IDs. This means the data is ready for basic analysis and modelling, but the text still needs cleaning before training a model.

2.2 Exploratory Data Analysis

The labels are not evenly spread across the dataset. Most comments are clean. Some labels are common, while others are rare. The most common label is toxic. The rarest label is threat.

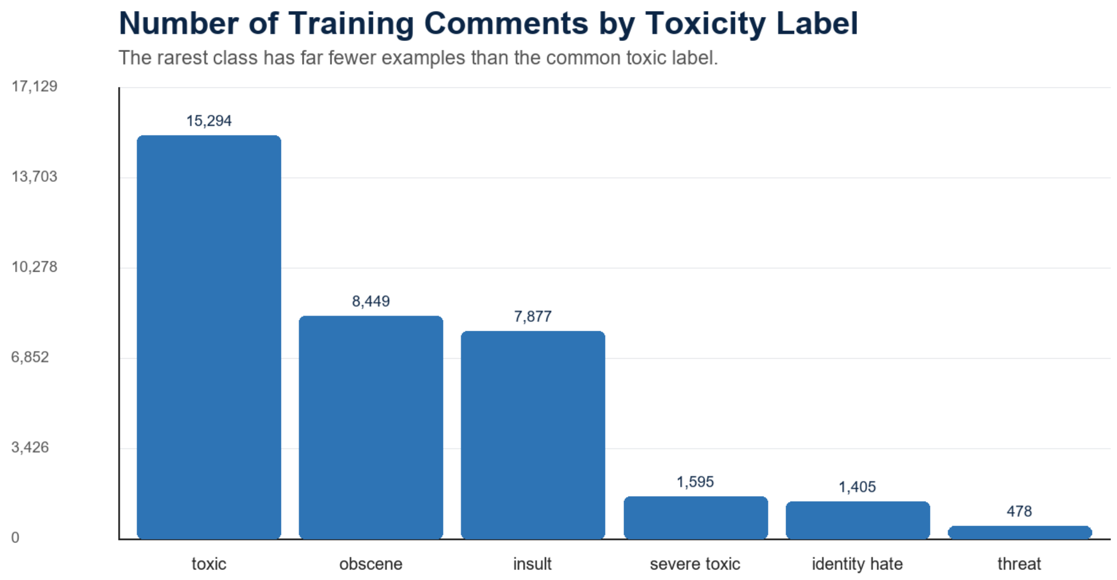

Figure 1. Number of training comments for each toxicity label.

Figure 1 visually shows another imbalance in the data set. The toxic label is the highest category of the data, and threats are much smaller in the data set. It is very important to note that when a label is rare, it gives the model fewer examples to learn from.

2.3 Clean and unclean comments.

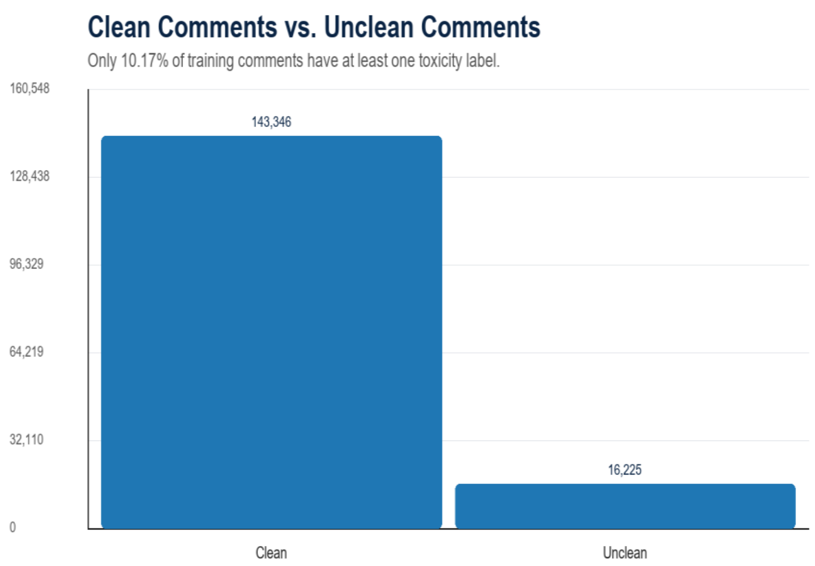

FIGURE 2. CLEAN VERSUS UNCLEAN COMMENTS IN THE TRAINING DATA.

Figure 2 shows the binary version of the class imbalance problem. A comment is treated as unclean if at least one of the six toxicity labels is positive. That means some comments can have more than one unclean label. version of the problem. A comment was treated as unclean if at least one of the six toxicity labels was active. By that definition, the training data contained 143,346 clean comments and 16,225 unclean comments. In percentage terms, 89.83% of the training comments were clean, and 10.17% were unclean. This confirms that the binary task is also imbalanced. The model should be carefully evaluated to ensure it detects toxic comments and does not simply predict "clean" for most rows.

The class distribution was analysed by counting how many comments belonged to each toxicity label. The analysis showed that the dataset is highly imbalanced, with most comments being clean and some labels, such as threat, identity_hate, and severe_toxic, appearing much less often. Because the data is a multilabeled data set and it is highly imbalanced, **Accuracy is not a good main metric** because most of the comments are clean. **Precision alone** can hide cases where the model misses many toxic comments. **Recall alone** can hide cases where the model labels too many clean comments as toxic. **F1-score** is better because it balances precision and recall.

3.0 Methodology

For each task, the project trained one model from scratch and one transfer-learning model. This produced four main experiments: a from-scratch neural network for multi-label classification, a from-scratch neural network for binary classification, a DistilBERT model for multi-label classification, and a DistilBERT model for binary classification. The figure below gives an illustration of the methodology.

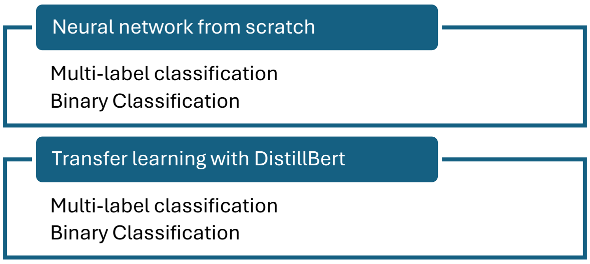

Figure 3. Methodology Illustration

3.1 Text processing

Text processing was carried out before training the two models in different ways because the two models require different text input formats. For neuro network, the cleaning of the raw text was carried out manually, and the text was converted into lowercase so that words such as “Bad” and “bad” are treated equally. Unnecessary Characters were removed, and the text was simplified to keep only letters, numbers and spaces. After that, each comment is split into individual words. A vocabulary was then built from the training data, and then each comment word was converted into IDs, and then the sequences were padded to a truncated length.

For DistilBERT, the pretrained DistilBERT tokeniser was used instead of manual word tokenization. It converted comments into input_ids and attention_mask, with padding and truncation to a maximum length of 128 tokens. This allowed the text to match the format expected by the pretrained model.

3.2 Evaluation metrics

The project used several metrics because no single number explains model quality on an imbalanced dataset. ROC-AUC measures ranking quality, it checks whether positive examples usually receive higher model scores than negative examples. Precision answers the question: when the model predicts a label, how often is it correct? Recall answers the question: among the true positive examples, how many did the model catch? F1 combines precision and recall into one score. F1 is especially useful here because both false positives and false negatives matter. A moderation system that flags too many clean comments wastes review effort and may frustrate users. A system that misses toxic comments fails to protect the discussion space.

Settings	Multiclass label from scratch	Binary from scratch
Vocabulary size	50,000	50,000
Embedding dimension	100	100

Hidden dimension	128	128
Dropout	0.3	0.3
Maximum length	212 words	200 words
Batch size	128	128
Epochs	10	10
Optimizer	Adam, learning rate 0.001	Adam, learning rate 0.001
Loss	BCEWithLogitsLoss with label pos_weight	BCEWithLogitsLoss with unclean pos_weight

TABLE 2. FROM-SCRATCH NEURAL NETWORK SETTINGS FROM THE COMPLETED NOTEBOOKS.

Transfer learning means the model starts from language knowledge learned before this project, then fine-tunes on the project dataset. This is useful for text because meaning often depends on context.

Setting	Multi-label DistilBERT	Binary DistilBERT
Base model	distilbert-base-uncased	distilbert-base-uncased
Outputs	6 labels	1 unclean score
Maximum token length	128	128
Batch size	16 train / 16 eval	16 train / 16 eval
Epochs	2	2
Learning rate	2e-5	2e-5
Weight decay	0.01	0.01
Best-model metric	ROC-AUC	ROC-AUC

TABLE 3. TRANSFER-LEARNING SETTINGS FROM THE COMPLETED NOTEBOOKS.

4.0 Multi-label Classification

The main modelling task is multi-label classification. This means that one comment can belong to more than one toxicity category at the same time. For example, one comment can be both toxic and insult, or toxic, obscene, and identity_hate together. Therefore, the model must make six separate yes or no decisions for each comment.

This is different from multi-class classification. In multi-class classification, only one class is selected. In this project, the six labels are independent outputs, so the output layer has six neurons. Each neuron produces one score for one label. During evaluation, sigmoid is used to convert each score into a probability. A threshold is then used to decide whether each label is predicted as present or absent.

4.1 Neural Network from Scratch

The first multi label model was a simple neural network built in PyTorch. Before training, the comments were cleaned by converting them to lowercase, then keeping only letters, numbers, and spaces, and removing extra spaces. The model had a vocabulary size of 50,000 words. Although this covers only part of the unique words, it covers 98.38% of all word in the training data, so most of the actual text tokens are represented.

Each comment was converted into word IDs and padded or shortened to a fixed length. The final notebook selected a maximum sequence length of 212 words, which covers about 94% of the training comments. Padding was used for shorter comments, while longer comments were shortened to this maximum length

The model architecture was following: Input -> Embedding -> Average Pooling -> Dense layer with ReLU -> Dropout -> Output logits. Average pooling combines all word vector into one sentence vector. This makes the model simple and efficient, but it also means that word order and detailed context are not wholly preserved.

The model was trained using BCEWithLogitsLoss, which is suitable for multi label classification because it treats every label as a separate binary decision. Then, Class weights were used to give more importance to rare labels such as threat, severe toxic, and identity hate. We used Adam optimizer with a learning rate of 0.001, batch size 128, and 10 epochs.

4.1.1 Neural Network from Scratch Test Results

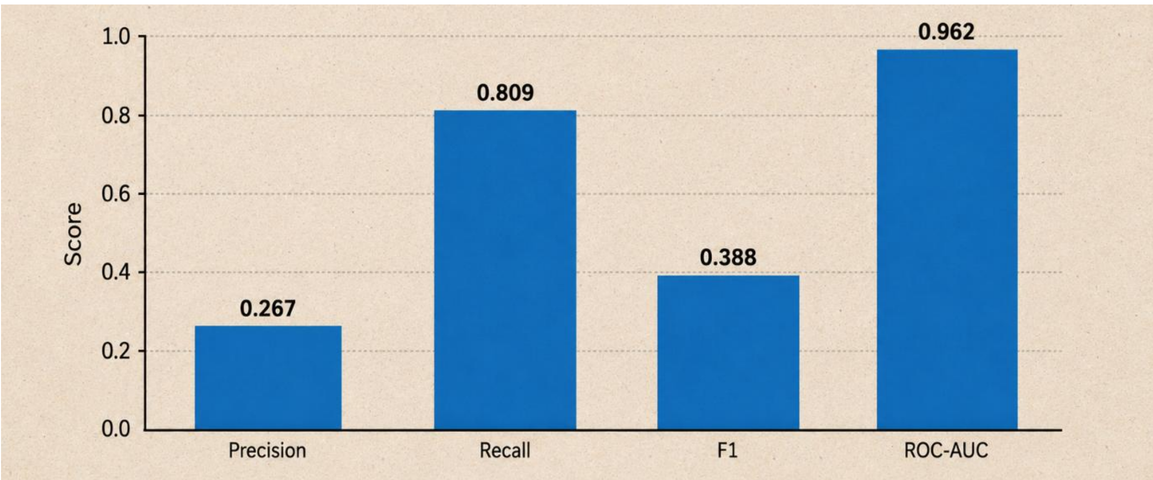

FIGURE 4. OVERALL RESULTS (NN FROM SCRATCH)

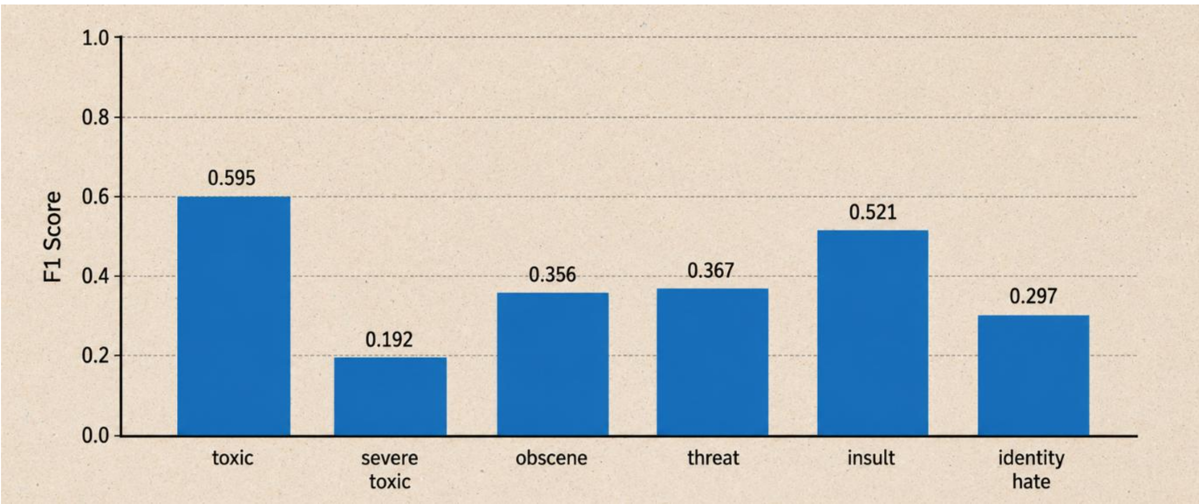

FIGURE 5. RESULTS BY COMPARING F1 FOR EACH LABEL (NN FROM SCRATCH)

The neural network achieved a good AUC, indicating its ability to learn useful features to distinguish the toxic and non-toxic labels. But the precision for some labels is low, especially for severe toxic and identity hate. This indicates that the model predicts these rare labels too often. The recall, on the other hand, is high, so the model is able to detect many real toxic comments but also produces many false positives.

4.2 Transfer Learning with DistilBERT

The second multi-label model used transfer learning with DistilBERT. Transfer learning starts from a model that has already learned general language patterns from a large amount of text. The pretrained model was then fine-tuned on the toxic comment dataset. This helps the model understand word order and context better than the simple average pooling neural network.

The model used distilbert-base-uncased with a classification head for six labels. The Hugging Face tokenizer converted comments into token IDs, with truncation and padding to a maximum length of 128 tokens. The model was loaded with 6 labels and problem_type="multi_label_classification", so the output is six independent logits.

The model outputs logits during training and sigmoid is used to convert the logits into probabilities. The validation set was also used to find a separate prediction threshold for each label. This is useful because rare labels often need different thresholds from common labels.

Label	Best threshold	Validation F1 at threshold
toxic	0.45	0.825
severe_toxic	0.21	0.524
obscene	0.32	0.829
threat	0.54	0.592
insult	0.32	0.766
identity_hate	0.46	0.577

Table 4. DistilBERT validation thresholds by label.

4.2.1 Transfer Learning Test Results

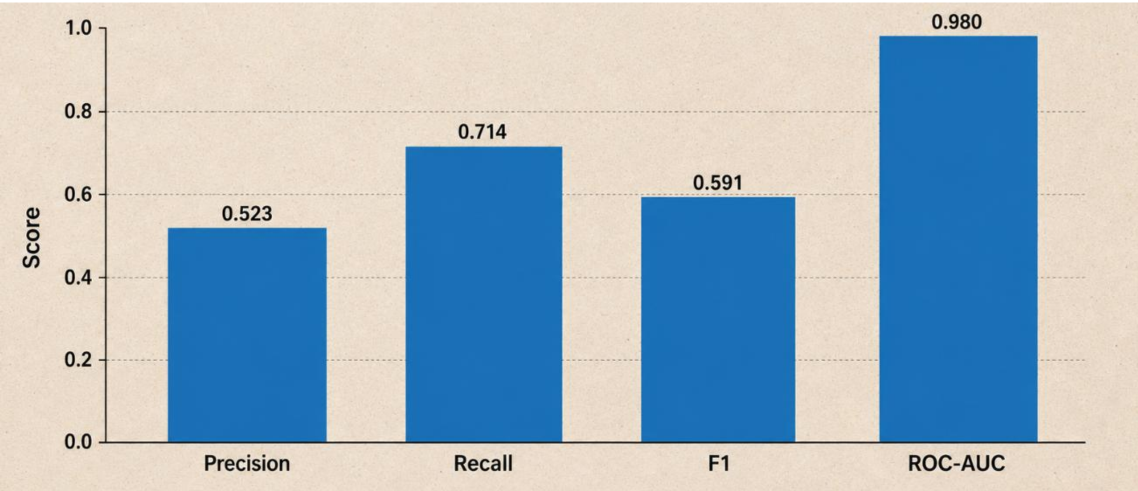

Figure 6. Overall Results (Transfer Learning)

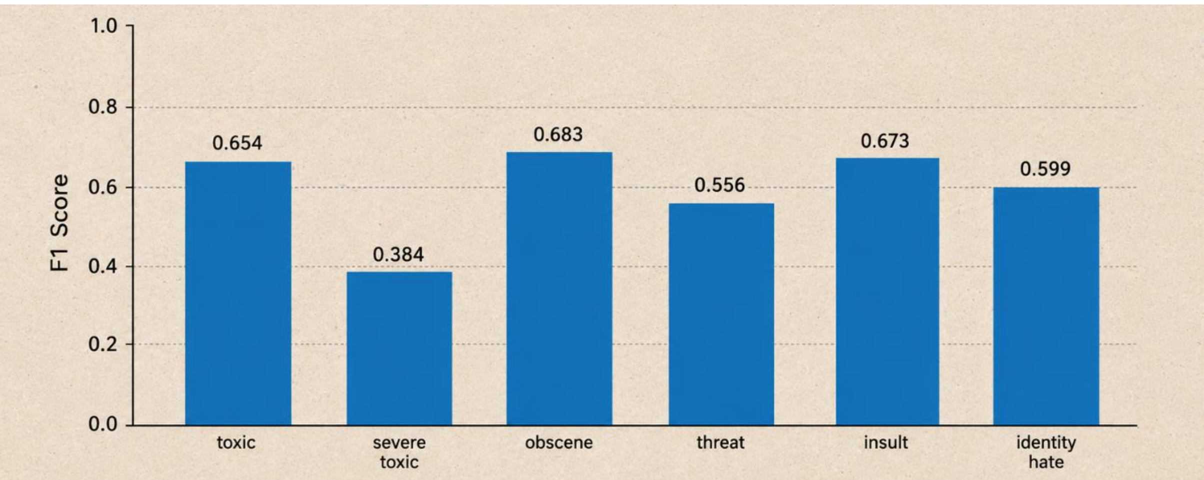

FIG 7. Results by comparing F1 for each label (Transfer Learning)

The transfer learning model performs better because DistilBERT can use contextual information. For example, it can consider the order of words and the surrounding meaning, while the simple neural network mainly averages word vectors. This is especially helpful for difficult and rare labels such as threat and identity_hate.

4.3 Comparison of the Two Multi-label Models

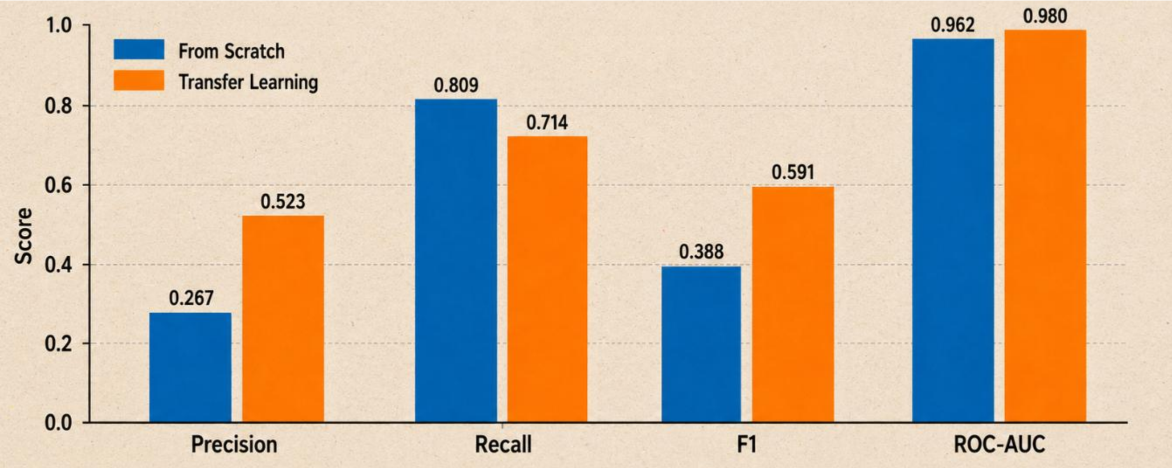

FIGURE 8. MULTI-LABEL OVERALL COMPARISON (SCRATCH VS TRANSFER LEARNING)

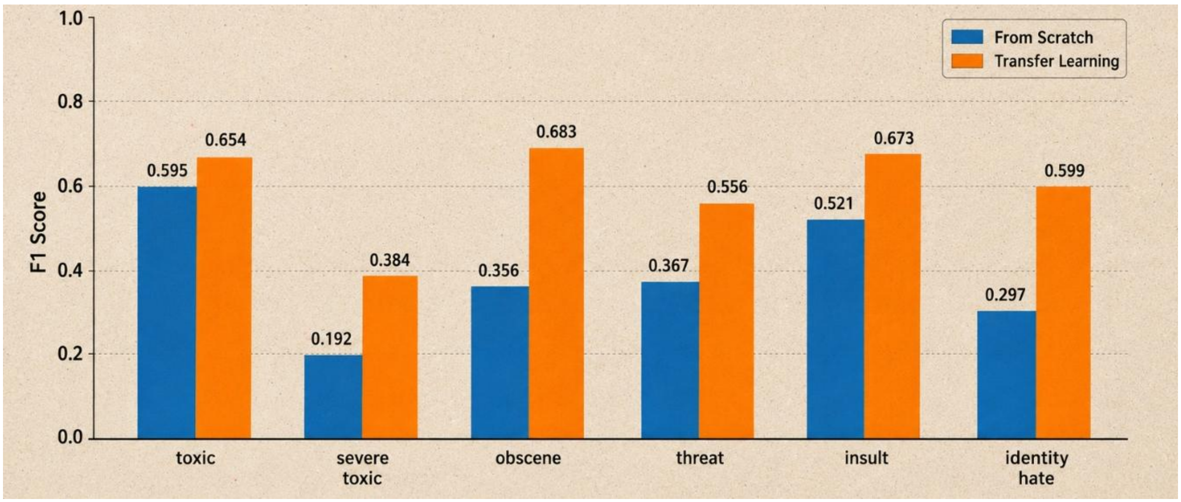

FIGURE 9. MULTI-LABEL COMPARISON BY F1 SCORE (SCRATCH VS TRANSFER LEARNING)

The comparison shows that the neural network from scratch has higher recall, meaning it finds many true toxic labels. However, its precision is much lower, so it also creates many false positives. The transfer learning model gives a better balance. It improves AUC, precision, and Macro F1, which makes it the stronger model overall.

5.0 Binary Classification

The second task is to compare the binary classification of a neural network from scratch with the binary classification of the transfer learning model. The binary classification task turns the data set with unclean labels into a single class. If a comment has at least one unclean label, it is considered unclean. The binary classification uses two models as described in the methodology, one neural network from scratch and then a transfer learning model with DistilBert. The original training data was split into training and validation, with 80% training and 20% validation. Because of this imbalance, accuracy alone was not enough to judge performance. A model could achieve high accuracy by mostly predicting clean, while still failing to detect harmful comments. Therefore, precision, recall, F1-score, ROC-AUC, and the confusion matrix were also used.

5.1 Binary neural network from scratch

The binary from-scratch model used the same general architecture but produced one output and the probability that a comment was unclean. Since the dataset contained many more clean comments than unclean comments, BCEWithLogitsLoss was used with a positive class weight to give more importance to the minority unclean class. The model was trained for 10 epochs using Adam with a learning rate of 0.001. The best model was selected using validation ROC-AUC. The model first evaluated using the normal/default threshold of **0.50** during training. Then it searched for a better threshold on the validation set and selected **0.75** as the best threshold for F1-score.

Metrics	Values
Threshold	0.75
Test AUC	0.948
Test accuracy	0.912
Test precision	0.533
Test recall	0.808
Test F1	0.642

Table 5. Binary Neural Network from-scratch test results.

The confusion matrix shows how the model classified the Clean and Unclean samples. The values in the matrix show the number of correct and incorrect predictions for each class. This makes it easier to see where the model performs well and where it makes mistakes.

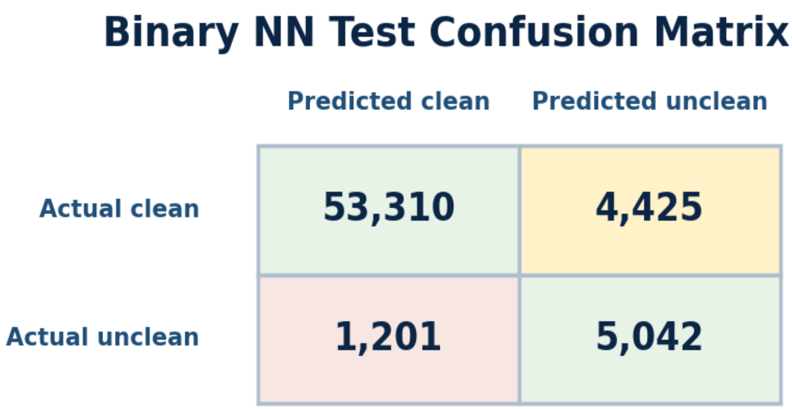

Figure 10. Binary Neural Network Confusion Matrix

Green cells are correct decisions. Yellow and red are review errors.

From the confusion matrix, we can see that overall, most of the predictions made by the model were correct from the test data. This means that the model performs well on the test data.

5.2 Binary DistilBERT

The second binary classification was done with DistilBERT it used transfer learning with distilbert-base-uncased. The tokenizer converted each comment into token IDs and attention masks, with a maximum sequence length of 128 tokens. The classification head was changed to produce one output logit, representing the probability that a comment was unclean.

Metric	Value
Threshold	0.86
Test ROC-AUC	0.973
Test accuracy	0.920
Test precision	0.554
Test recall	0.905
Test F1	0.687

*Table **5**. Binary **transfer learning** results.*

The notebook also first tested the default threshold of 0.50, but the final reported DistilBERT binary result should use 0.86, because that threshold was selected from the validation set to improve the F1-score

Binary Transfer Learning Confusion Matrix

	Predicted clean	Predicted unclean
Actual clean	53,196	4,539
Actual unclean	596	5,647

Figure 11. Binary Transfer Learning Confusion Matrix

Green cells are correct decisions. Yellow and red are review errors.

From the confusion matrix, the model correctly classified 53,196 clean comments and 5,647 unclean comments, while 4,539 clean comments were incorrectly flagged as unclean and 596 unclean comments were missed.

Metric	Binary NN From Scratch	Binary DistilBERT
Threshold	0.75	0.86
Test ROC-AUC	0.948	0.973
Test accuracy	0.912	0.920
Test precision	0.533	0.554
Test recall	0.808	0.905
Test F1-score	0.642	0.687

*Table **5**. Binary test results comparison.*

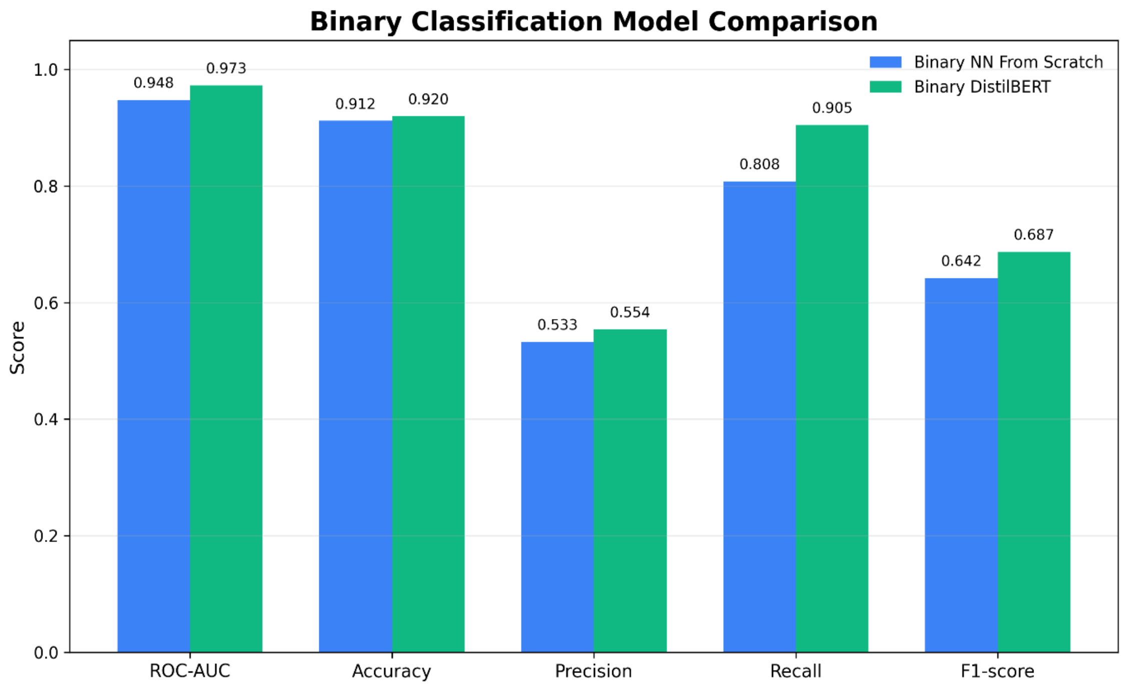

Figure 12. Binary Models metrics comparison

From the table above, the binary DistilBERT model outperformed the neural network from scratch across all main evaluation metrics. It achieved a higher ROC-AUC, accuracy, precision, recall, and F1-score. The improvement in the recall is important, as DistilBERT missed only 596 unclean comments compared with 1,201 missed by the neural network. This shows that the pretrained language model was better at detecting toxic or unclean comments.

6.0 Conclusion

Machine learning can identify toxic comments with useful accuracy, but the best results came from transfer learning. The multi-label neural network learned useful patterns across the six toxicity categories, but it struggled more with rare labels. The multi-label DistilBERT model achieved the highest overall ROC-AUC, showing the strongest ranking ability across the six toxicity labels. The binary DistilBERT model performed best on the clean-versus-unclean task, giving the highest binary F1-score and recall. Overall, DistilBERT improved both the multi-label and binary tasks compared with the from-scratch neural networks. In a practical application, machine learning models should be used to support human review, as false positives and false negatives remain.

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